Schmidt Orthogonalisation Process

$S = \{1, t, t^2\}$ on $[-1, 1]$

$$\langle f, g \rangle = \int_a^b f(t)\, g(t)\, dt$$

$$= \int_{-1}^{1} f(t)\, g(t)\, dt$$

$w_1 = f_1 = 1$

$$\Rightarrow w_2 = f_2 - \frac{\langle f_2, w_1 \rangle}{\langle w_1, w_1 \rangle} w_1 = t \quad , \langle f_2, w_1 \rangle = \int_{-1}^{1} t\, dt = 0$$

$$w_3 = f_3 - \frac{\langle f_3, w_1 \rangle}{\langle w_1, w_1 \rangle} w_1 - \frac{\langle f_3, w_2 \rangle}{\langle w_2, w_2 \rangle} w_2$$

$$\langle w_1, w_1 \rangle = \int_{-1}^{1} dt = 2 .$$

$$\langle f_3, w_1 \rangle = \int_{-1}^{1} t^2\, dt = \left[\frac{t^3}{3}\right]_{-1}^{1} = \frac{2}{3} .$$

$$\langle f_3, w_2 \rangle = \int_{-1}^{1} t^3\, dt = 0 .$$

Thm:-

Let $\{v_1, v_2, \ldots, v_n\}$ be any basis of an inner product space V. Then $\exists$ an orthonormal basis $\{u_1, u_2, \ldots, u_n\}$ of V such that the change of basis matrix from $\{v_i\}$ to $\{u_i\}$ is triangular.

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